

Why Warmup Helps: A Non-Asymptotic Analysis of the WSD Schedule under (H_0, H_1) -smoothness and Aiming Condition

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Aug 25, 2026

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Highlights

Focus: optimization viewpoint for the special class of functions. Justifications of the assumptions are mostly compensated by previous works.

- Rigorous linear warmup analysis and fair comparison with constant stepsize

- Clear development of theory of SGD inspired by noiseless GD settings

- Relaxation of assumptions, e.g. interpolation condition, convexity (caveat: we have assumed properties of the loss landscape that can be viewed as a case of convexity), full knowledge about the value of f^* and f_{S_k}

- One reason for lowering the peak learning rate under longer training horizon: bounding noise amplification to keep iterates around the basin

- Plug-in-and-play decaying stepsizes

- Self-tuning warmup strategy theoretically (much) better (Experiments: To do!)

- Quantification of warmup shapes and stage partitioning, connecting theory with practice

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- 3 Uniform Localization under Mean-Zero Noise: Analysis of SGD
- 4 Self-Tuning Schedules

Setup and Notations

Assumptions

Assumption 1 (Generalized Smoothness)

The function $f : \mathbb{R}^d \rightarrow \mathbb{R}$ with $f^* := \inf_{\mathbf{w} \in \mathbb{R}^d} f(\mathbf{w}) > -\infty$ is twice differentiable and (ρ, H_0, H_ρ) -smooth, i.e.

$$\|\nabla^2 f(\mathbf{w})\| \leq H_0 + H_\rho(f(\mathbf{w}) - f^*)^\rho, \quad H_0, H_\rho \geq 0, \rho > 0.$$

Throughout we take $\rho = 1$ and write $H_1 := H_\rho$.

Assumption 2 (Aiming Condition)

There is $\theta > 0$ such that, with $\mathcal{S} := \arg \min_{\mathbf{w} \in \mathbb{R}^d} f(\mathbf{w})$,

$$\langle \nabla f(\mathbf{w}), \mathbf{w} - \pi_{\mathcal{S}}(\mathbf{w}) \rangle \geq \theta(f(\mathbf{w}) - f^*) \quad \forall \mathbf{w} \in \mathbb{R}^d.$$

Assumption 3 (Sub-Gaussian Noise)

The stochastic noise ξ_k has $\mathbb{E}[\xi_k \mid \mathcal{F}_k] = 0$ and

$$\mathbb{P}(\|\xi_k\| \geq t \mid \mathcal{F}_k) \leq 2 \exp(-t^2/(2\sigma^2)), \quad t \geq 0,$$

where $\mathcal{F}_k := \sigma(\mathbf{w}_0, \dots, \mathbf{w}_k)$.

Definition 1 (WSD Schedule)

Given $T := T_w + T_s + T_d$ (warmup, stable, decay), a monotonically decreasing $\psi : [0, 1] \rightarrow [0, 1]$ with $\psi(0) = 1$, $\psi(1) = 0$, and $0 < \eta_{\text{start}} \leq \eta_{\text{peak}}$, set

$$\eta_k := \begin{cases} \eta_{\text{start}} + \frac{k}{T_w}(\eta_{\text{peak}} - \eta_{\text{start}}), & 0 \leq k < T_w, \\ \eta_{\text{peak}}, & T_w \leq k < T_w + T_s, \\ \eta_{\text{peak}}\psi\left(\frac{k - T_w - T_s}{T_d}\right), & T_w + T_s \leq k \leq T. \end{cases}$$

Moreover, we write $S_k := \sum_{j < k} \eta_j$.

Warmup Provably Helps: Analysis of GD

Analysis of GD

We consider the deterministic gradient descent algorithm, where

$$\mathbf{w}_{k+1} = \mathbf{w}_k - \eta_k \nabla f(\mathbf{w}_k), \Delta_k := f(\mathbf{w}_k) - f^*, D_k := \text{dist}(\mathbf{w}_k, \mathcal{S}) = \|\mathbf{w}_k - \pi_{\mathcal{S}}(\mathbf{w}_k)\|.$$

Theorem 1 (GD Post-Warmup Last-Iterate Guarantees)

*Suppose that Assumptions 1–2 holds. Let $U \geq \Delta_0$ be any **upper bound** on the initial suboptimality gap, $c \in (0, 1/3]$ be a constant and $\{\mathbf{w}_k\}$ be generated by GD, with the learning-rate schedule of Definition 1, where*

$$\eta_{\text{start}} := \frac{c}{2(H_0 + H_1 U)}, \quad \eta_{\text{peak}} := \frac{c}{2H_0}, \quad \tilde{D} := \sqrt{D_0^2 + \frac{cU}{2(1-c)H_0}},$$

and

$$T_w \geq \left\lceil 1 + \frac{H_1 U}{H_0} \right\rceil \cdot \max \left\{ \left\lceil \frac{36 H_1 \tilde{D}^2}{c \theta^2} \right\rceil, \left\lceil \frac{24}{c(1-c)} \right\rceil \right\}.$$

Analysis of GD

Theorem 1 (GD Post-Warmup Last-Iterate Guarantees)

With $\alpha := (1 - c) \cdot \min\{\theta^2/\tilde{D}^2, H_1\}$, $\Delta_{T_w} \leq H_0/(3H_1)$, and for every $T_w < k \leq T_w + T_s$, we have

$$\Delta_k \leq \frac{2H_0}{\alpha c(k - T_w)}, \quad \|\nabla f(\mathbf{w}_k)\|^2 \leq \frac{9H_0^2}{2\alpha c(k - T_w)} + \frac{27H_0^2 H_1}{\alpha^2 c^2 (k - T_w)^2};$$

In contrast, for the iterates $\{\mathbf{w}_k\}$ generated by constant-stepsize GD with the largest admissible step $\eta_c = \eta_{\text{start}}$, by setting $\tilde{D}_c := \sqrt{D_0^2 + \eta_c U/(1 - c)} \leq \tilde{D}$ and

$\alpha_c := (1 - c) \min\{\theta^2/\tilde{D}_c^2, H_1\} \geq \alpha$, we have the guarantee that

$$\Delta_k \leq \frac{2(H_0 + H_1 U)}{\alpha_c c k}, \quad \|\nabla f(\mathbf{w}_k)\|^2 \leq \frac{9H_0(H_0 + H_1 U)}{2\alpha_c c k} + \frac{27H_1(H_0 + H_1 U)^2}{\alpha_c^2 c^2 k^2},$$

for all $k \geq 1$.

Spending Negligible Iterations on Warmup Is Indeed Better

Remark 1 (Last-Iterate Comparison)

From Theorem 1, we can conduct a comparison of the last-iterate bounds. For $k \gg T_w$, the guarantees provided by the two schedules differ only in H_0 versus $H_0 + H_1\Delta_0$ and in α versus α_c . We consider the suboptimality gap bounds and the leading $1/k$ term of the squared gradient bounds, where we have

$$\frac{\Delta_k^{const}}{\Delta_k^{warmup}}, \frac{\|\nabla f(\mathbf{w}_k)\|_{const}^2}{\|\nabla f(\mathbf{w}_k)\|_{warmup}^2} \rightarrow \frac{(H_0 + H_1\Delta_0)\alpha}{H_0\alpha_c} = \left(1 + \frac{H_1\Delta_0}{H_0}\right) \frac{\alpha}{\alpha_c} \geq 1.$$

Warmup schedule is always no worse than the constant-stepsizes schedule, with respect to the ratio above.

Uniform Localization under Mean-Zero Noise: Analysis of SGD

Analysis of SGD

Assumption 4 (Stage Lengths)

Let $U \geq \Delta_0$ be any **upper bound** on the initial suboptimality gap. Suppose Assumptions 1–3 holds. Let $c \in (0, 1/36]$ be a constant, $\delta \in (0, 1)$ be the failure probability, $\Delta_0 := f(\mathbf{w}_0) - f^*$, and $\sigma_\delta := \sigma \sqrt{2 \ln(4T/\delta)}$. Using the WSD schedule as in Definition 1 and fix the constants

$$\eta_{\text{start}} := \min \left\{ \frac{c}{2(H_0 + H_1 U)}, \frac{c}{2M} \right\}, \quad \eta_{\text{peak}} := \min \left\{ \frac{c}{2H_0}, \frac{c}{2M} \right\},$$

$$\alpha := \frac{1 - 2c}{2} \min \left\{ \frac{\theta^2}{4\tilde{D}^2}, H_1 \right\},$$

$$M := 4\sigma_\delta H_1 \max \left\{ \frac{2\tilde{D}}{\theta}, \frac{1}{\sqrt{H_1}} \right\}, \quad \tilde{D} := \sqrt{D_0^2 + \frac{U}{4H_0}},$$

$$\Delta_{\text{th}} := \sqrt{\frac{1 + 4c(1 - 2c)}{(1 - 2c)^2}} \cdot \sigma_\delta \cdot \max \left\{ \frac{2\tilde{D}}{\theta}, \frac{1}{\sqrt{H_1}} \right\}.$$

Analysis of SGD

Assumption 4 (Stage Lengths)

We require the stage lengths to satisfy

$$T_w \geq \max \left\{ 2, \min \left\{ \left\lceil \frac{64H_1}{\alpha c} \right\rceil, \left\lceil \frac{H_1 \tilde{D}^2}{\theta^2} \right\rceil \right\} \right\} \cdot \left\lceil \frac{H_0 + H_1 U}{M} \right\rceil.$$

Theorem 2 (Convergence Properties of the WSD Schedule with Mean-Zero Noise)

Fix $c \in (0, 1/36]$, $\delta \in (0, 1)$, and suppose that Assumptions 1–3 holds. Let $\{\mathbf{w}_k\}$ be generated by SGD, with the learning-rate schedule of Definition 1 satisfying Assumption 4 and

$$V_T := \sum_{k < T} \eta_k^2 \leq \frac{\tilde{D}^2}{1536\sigma_\delta^2 \ln(2/\delta)}.$$

Analysis of SGD

Theorem 2 (Convergence Properties of the WSD Schedule with Mean-Zero Noise)

Then, there is an event $\mathcal{E}$ with $\mathbb{P}(\mathcal{E}) \geq 1 - \delta$ such that $\Delta_{T_w} \leq \max\{2\Delta_{\text{th}}, H_0/H_1\}$, the minimum gradient satisfies

$$\mathbb{E} \left[\min_{T_w \leq k \leq T} \|\nabla f(\mathbf{w}_k)\|^2 \middle| \mathcal{E} \right] \leq \frac{1}{(1-c)\mathbb{P}(\mathcal{E})} \left(\frac{\mathbb{E}[\Delta_{T_w}]}{\sum_{k=T_w}^{T-1} \eta_k} + (H_0 + 2H_1\Delta_{\text{th}})\sigma_\delta^2 \frac{\sum_{k=T_w}^{T-1} \eta_k^2}{\sum_{k=T_w}^{T-1} \eta_k} \right);$$

Moreover, the function values satisfy

$$\frac{\sum_{k=T_w}^{T-1} \eta_k \Delta_k}{\sum_{k=T_w}^{T-1} \eta_k} \leq \frac{5\tilde{D}^2 + \frac{8}{1-2c}\eta_{\text{peak}}\Delta_{\text{th}}}{2\theta \sum_{k=T_w}^{T-1} \eta_k}.$$

Analysis of SGD

Theorem 2 (Convergence Properties of the WSD Schedule with Mean-Zero Noise)

Furthermore, using the decay profile $\psi(x) := 1/\sqrt{1+xT_d}$, so that for k in the decay phase $\eta_k = \eta_{\text{peak}}(1 + (k - T_w - T_s))^{-1/2}$, we have that

$$\mathbb{E} \left[\min_{T_w \leq k \leq T} \|\nabla f(\mathbf{w}_k)\|^2 \middle| \mathcal{E} \right] \leq \frac{1}{(1-c)\mathbb{P}(\mathcal{E})} \left(\frac{2\Delta_{\text{th}}}{\eta_{\text{peak}}(T_s + \sqrt{T_d})} + \frac{(H_0 + 2H_1\Delta_{\text{th}})\sigma_\delta^2\eta_{\text{peak}}(T_s + 1 + \ln T_d)}{T_s + \sqrt{T_d}} \right),$$

$$\frac{\sum_{k=T_w}^{T-1} \eta_k \Delta_k}{\sum_{k=T_w}^{T-1} \eta_k} \leq \frac{1}{2\theta(T_s + \sqrt{T_d})} \cdot \left(\frac{5\tilde{D}^2}{\eta_{\text{peak}}} + \frac{8\Delta_{\text{th}}}{1-2c} \right).$$

Iteration Complexities

Remark 2 (Explicit Regimes for Fully-Resolved Iteration Complexities)

Let T_w be the number of iterations for the suboptimality gap to reach the noise floor, T_d^{grad} be the number of iterations to guarantee that $\mathbb{E}[\min_{T_w \leq k < T} \|\nabla f(\mathbf{w}_k)\|^2 \mid \mathcal{E}] \leq \varepsilon_g^2$ and T_d^{subopt} be the number of iterations to guarantee that $\sum_{k=T_w}^{T-1} \eta_k \Delta_k / \sum_{k=T_w}^{T-1} \eta_k \leq \varepsilon_s$.

We resolve the min and max operators by considering the two branches of the contraction coefficient α and the two branches of the peak-step cap η_{peak} in the iteration complexity bounds and provide explicit stage-length formulas for four regimes as follows:

Contraction branches. From $\alpha = (1 - 2c) \min\{\theta^2/(4\tilde{D}^2), H_1\}$, we consider two conditions: $\theta^2/(4\tilde{D}^2) \leq H_1$ (A1, aiming-limited) and $\theta^2/(4\tilde{D}^2) > H_1$ (A2, smoothness-limited).

Peak-step branches. From $\eta_{\text{peak}} = c/(2 \max\{H_0, M\})$, we also have two conditions: $H_0 \geq M$ (E1, smoothness-capped), i.e. $\eta_{\text{peak}} = c/(2H_0)$, and $H_0 < M$ (E2, noise-capped), i.e. $\eta_{\text{peak}} = c/(2M)$.

Iteration Complexities

Remark 2 (Explicit Regimes for Fully-Resolved Iteration Complexities (cont.))

Regime	Conditions	Warmup T_w	Min. Gradient T_d^{grad}	Weighted Subopt. T_d^{subopt}
1	$H_0 \geq \frac{8\sigma_\delta H_1 \tilde{D}}{\theta}, \theta \leq 2\tilde{D}\sqrt{H_1}$	$\tilde{\Theta}\left(\frac{(H_0+H_1\Delta_0)\tilde{D}^2}{\sigma_\delta H_1 \theta^2}\right)$	$\tilde{\Theta}\left(\frac{H_0^2 \tilde{D}^2 \sigma_\delta^2}{\theta^2 \varepsilon_g^4 (1-\delta)^2}\right)$	$\tilde{\Theta}\left(\frac{H_0^2 \tilde{D}^4}{\theta^2 \varepsilon_s^2}\right)$
2	$H_0 < \frac{8\sigma_\delta H_1 \tilde{D}}{\theta}, \theta \leq 2\tilde{D}\sqrt{H_1}$	$\tilde{\Theta}\left(\frac{(H_0+H_1\Delta_0)\tilde{D}^2}{\sigma_\delta H_1 \theta^2}\right)$	$\tilde{\Theta}\left(\frac{H_1^2 \tilde{D}^4 \sigma_\delta^4}{\theta^4 \varepsilon_g^4 (1-\delta)^2}\right)$	$\tilde{\Theta}\left(\frac{H_1^2 \tilde{D}^6 \sigma_\delta^2}{\theta^4 \varepsilon_s^2}\right)$
3	$H_0 \geq 4\sigma_\delta \sqrt{H_1}, \theta > 2\tilde{D}\sqrt{H_1}$	$\tilde{\Theta}\left(\frac{H_0+H_1\Delta_0}{\sigma_\delta \sqrt{H_1}}\right)$	$\tilde{\Theta}\left(\frac{H_0^2 \sigma_\delta^2}{H_1 \varepsilon_g^4 (1-\delta)^2}\right)$	$\tilde{\Theta}\left(\frac{H_0^2 \tilde{D}^4}{\theta^2 \varepsilon_s^2}\right)$
4	$H_0 < 4\sigma_\delta \sqrt{H_1}, \theta > 2\tilde{D}\sqrt{H_1}$	$\tilde{\Theta}\left(\frac{H_0+H_1\Delta_0}{\sigma_\delta \sqrt{H_1}}\right)$	$\tilde{\Theta}\left(\frac{\sigma_\delta^4}{\varepsilon_g^4 (1-\delta)^2}\right)$	$\tilde{\Theta}\left(\frac{H_1 \tilde{D}^4 \sigma_\delta^2}{\theta^2 \varepsilon_s^2}\right)$

Table: Stage lengths by regime.

Self-Tuning Schedules

Self-Tuning Schedules: GD

Definition 2 (Saturating Warmup for GD)

Suppose that Assumptions 1–2 hold. Let $U \geq \Delta_0$ be any upper bound on the initial suboptimality gap and let $c \in (0, 1/3]$, $\alpha = (1 - c) \min\{\theta^2/\bar{D}_0^2, H_1\}$, where $\tilde{D} := \sqrt{D_0^2 + \frac{cU}{(1-c)H_0}}$. Set $\hat{\Delta}_0 := U$ and for all $k \geq 0$, let

$$\eta_k := \frac{c}{H_0 + H_1 \hat{\Delta}_k}, \quad \hat{\Delta}_{k+1} := \frac{\hat{\Delta}_k}{1 + \alpha \eta_k \hat{\Delta}_k}.$$

Theorem 3 (Guarantees of GD under the Self-Tuning Warmup Schedule)

Suppose Assumptions 1–2 hold and fix $c \in (0, 1/3]$. Let $\{\mathbf{w}_k\}$ be generated by GD using the schedule as in Definition 2, the suboptimality gap reaches the post-warmup benign-curvature state $\hat{\Delta}_k \leq H_0/H_1$ after

$$T_w \geq \max \left\{ 0, \left\lceil 1 + \ln \left(\frac{H_1 U}{H_0} \right) \right\rceil \cdot \max \left\{ \left\lceil \frac{1}{2c(1-c)} \right\rceil, \left\lceil \frac{H_1 \tilde{D}^2}{2c(1-c)\theta^2} \right\rceil \right\} \right\}$$

steps, which is logarithmic in $H_1 U/H_0$ compared to Theorem 1.

Self-Tuning Schedules: SGD

Definition 3 (Saturating Warmup for SGD)

Suppose that Assumptions 1–3 hold. Let $U \geq \Delta_0$ be any upper bound on the initial suboptimality gap and let $c \in (0, 1/36]$,

$$\alpha := \frac{1-2c}{2} \min \left\{ \frac{\theta^2}{4\tilde{D}^2}, H_1 \right\}, \quad \tilde{D}_0 := \sqrt{D_0^2 + \frac{U}{4H_0}}.$$

Set $\hat{\Delta}_0 := U$ and for all $k \geq 0$, let

$$\eta_k := \frac{c}{H_0 + H_1 \hat{\Delta}_k}, \quad \hat{\Delta}_{k+1} := \max \left\{ \hat{\Delta}_k - \alpha \eta_k (\hat{\Delta}_k^2 - \Delta_{\text{th}}^2), \Delta_{\text{th}} \right\},$$

where

$$\Delta_{\text{th}} := \sqrt{\frac{1+4c(1-2c)}{(1-2c)^2}} \cdot \sigma_\delta \cdot \max \left\{ \frac{2\tilde{D}}{\theta}, \frac{1}{\sqrt{H_1}} \right\}.$$

f^* -Free Self-Tuning Schedules: SGD

Theorem 4 (Guarantees of SGD under the Self-Tuning Warmup Schedule)

Suppose Assumptions 1–2 hold and fix $c \in (0, 1/36]$. Let $\{\mathbf{w}_k\}$ be generated by SGD using the schedule as in Definition 3,

$$V_T := \sum_{k < T} \eta_k^2 \leq \frac{\tilde{D}^2}{1536\sigma_\delta^2 \ln(2/\delta)},$$

with

$$T_w \geq 1 + \left\lceil \frac{11}{6\alpha c} \cdot \left(\frac{H_0}{2\Delta_{\text{th}}} + H_1 \log \frac{U}{2\Delta_{\text{th}}} \right) \right\rceil = \mathcal{O} \left(\frac{1}{\alpha c} \cdot \left(\frac{H_0}{\sigma_\delta \cdot \alpha^{-1/2}} + H_1 \log \frac{U}{\sigma_\delta \alpha^{-1/2}} \right) \right),$$

where the saturating schedule achieves a logarithmic speedup, instead of linear as in Theorem 2, in the initial suboptimality gap estimate U again. Then, there is an event $\mathcal{E}$ with $\mathbb{P}(\mathcal{E}) \geq 1 - \delta$ such that $\Delta_{T_w} \leq 2\Delta_{\text{th}}$.

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